

## 02-ELECTRON DIFFRACTION

### PURPOSE of the EXPERIMENT

To observe the diffraction of free electrons through a polycrystalline graphite layer and prove that electrons have wave characteristics. To calculate the lattice constants  $d_1$  and  $d_2$  of graphite.

### GENERAL INFORMATION

In 1923, the French physicist L. de Broglie thought massive particles should also have wave behavior. According to Louis De Broglie's hypothesis, any particle with momentum  $p$ ,  $h=6.626 \cdot 10^{-34}$  js Planck's constant

$$\lambda = \frac{h}{p} \quad (1)$$

It also has the character of a wave with a wavelength. In 1927, Davisson and Germer experimentally confirmed the De Broglie hypothesis by showing that electrons diffract from nickel crystals. De Broglie was awarded the Nobel Prize in Physics in 1929.

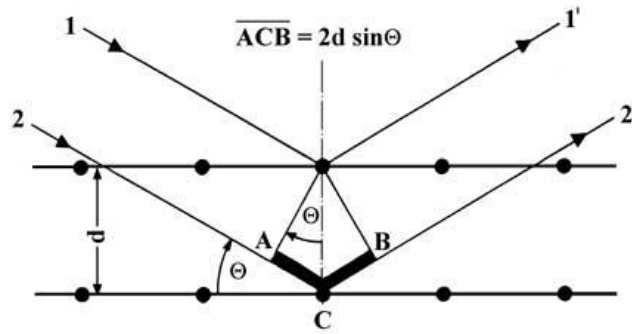
The de Broglie wavelength of an electron accelerated by the voltage  $V_a$  can be found;

$$KE = \frac{p^2}{2m} = eV_a \quad (2)$$
$$p = \sqrt{2emV_a} \Rightarrow \lambda = \frac{h}{\sqrt{2emV_a}}$$

An opaque obstacle, a slit, or a diffraction network can be used to observe the diffraction phenomenon. In this experiment, a diffraction grating obtained by sputtering carbon on a Ni (Nickel) grid was used as a target. 12 mg C contains  $6.1023$  atoms. Bragg diffraction occurs when waves are scattered from crystalline solids. The diffraction of electron waves sent onto the crystal structure is shown in Figure 1.

$$AC = d \sin \theta$$

$$ACB = 2d \sin \theta$$

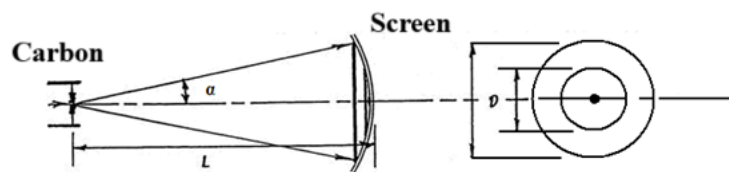


**Figure 1:** Bragg diffraction

Waves incident on a crystalline solid can be scattered from different planes in the crystal structure. This scattering from different layers causes a path difference between the waves. The path difference between waves scattered from neighboring planes is given by  $2d\sin\theta$ , where  $d$  is the distance between the planes. Bragg diffraction condition for the angle  $\theta$  that the reflected wave makes with the surface of the solid,

$$2d \sin \theta = n\lambda \quad ; \quad n = 0, 1, 2, \dots \quad (3)$$

If the electron behaves as a wave, a diffraction pattern should be observed on a screen  $L$  away from the crystal. Indeed, as can be seen in this experiment, two rings with a bright center are observed on the screen. The change in the anode voltage ( $V_a$ ), which accelerates the electrons, causes the radius to change. In other words, as the voltage decreases, the ring radius value increases diffraction increases. This result is in agreement with de Broglie's assumption. A decrease in voltage leads to an increase in wavelength, which leads to an increase in angular dispersion due to diffraction conditions.



**Figure 2:** The geometric shape of the interference pattern.

Electron waves from the crystal are scattered at an angle of  $2\theta$  with the direction of incidence of the electrons.

$$\tan 2\theta = \frac{R}{2L}$$

When we take into account the small-angle approach,

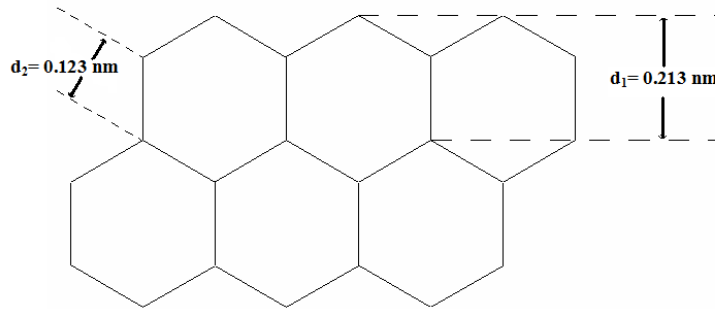
$$\lambda = d \frac{R}{2L} \quad (4)$$

$$R = \frac{2Lh}{d\sqrt{2meV_a}}$$

The radius of the glass bulb used in the experiment is 6.75 cm.

$$d = \frac{2Lh}{R\sqrt{2meV_a}} \quad (5)$$

Graphite crystals have a hexagonal geometry. Therefore, electron waves sent to graphite crystals will undergo Bragg diffraction from two different spaced planes. The width of these intervals is  $d_1 = 0.213$  nm and  $d_2 = 0.123$  nm as given in Figure 3.



**Figure 3:** Crystal structure of graphite.

In this case, for Bragg diffraction consisting of two separate gaps, two separate rings with diameters  $R_1$  and  $R_2$  will be formed. The crystal gaps caused by these rings are as follows.

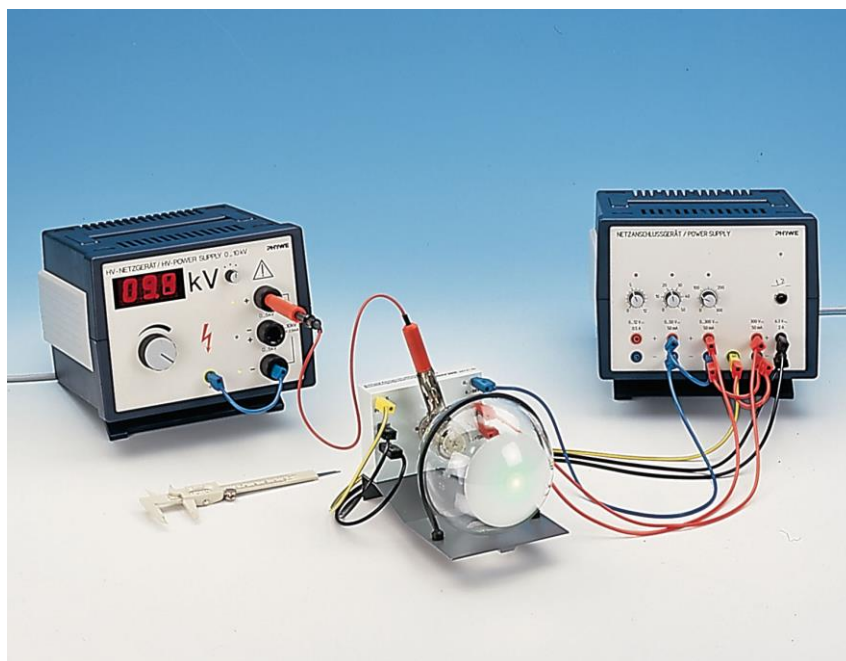
$$d_{10} = \frac{2Lh}{R_1\sqrt{2meV_a}} \quad d_{11} = \frac{2Lh}{R_2\sqrt{2meV_a}} \quad (6)$$

Here  $m$  and  $e$  are the mass and charge of the electron.

## HOW TO CONDUCT THE EXPERIMENT

Electron diffraction tube, high voltage source (0-10 kV), power supply (0-600 V)

The electron diffraction experimental setup in which diffraction will be observed is shown in Figure 4.



**Figure 4:** Experimental setup

The experimental setup consists of an electron diffraction tube and a high-voltage power supply. In the diffraction tube, electrons released from the electron gun are accelerated and pass through the polycrystalline graphite sample. Some of the accelerated electrons undergo Bragg diffraction from these crystals. The diffracted electrons produce a diffraction pattern on the fluorescence screen at the front of the tube.

1. Apply a voltage of 6V to the heater circuit. Wait 2 minutes for the circuit to fully warm up.
2. Apply a high voltage of 5kV to accelerate the electrons.
3. For the diffraction pattern you see on the screen of two luminous concentric rings, vary the voltage, measure the diameter of the rings for each voltage value, and record them in the table.
4. Calculate  $d_{10}$  and  $d_{11}$  from the values in the table and calculate the average value with error.

5. Plot  $\frac{1}{\sqrt{V_0}}$ -graphs separately with the diameter values of  $R_1$  and  $R_2$  found in step 3. Calculate the crystal spacings  $d_{10}$  and  $d_{11}$  by substituting the slope you found from these graphs in relation (6). Compare with the values found in step 4.

**Table 1:** Results

| V (kV) | $R_1$ (cm) | $R_2$ (cm) | $d_{10}$ | $d_{11}$ |
|--------|------------|------------|----------|----------|
|        |            |            |          |          |
|        |            |            |          |          |
|        |            |            |          |          |
|        |            |            |          |          |
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## REFERENCES

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3. [http://www.cce.ufes.br/jair/web/difr\\_el.pdf](http://www.cce.ufes.br/jair/web/difr_el.pdf)